Automatically coding economic activities via semantic search

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Abstract
Official economic activity classifications provide the reference framework for producing coherent and comparable statistics, yet their complexity makes accurate self-classification challenging. National Statistical Institutes (NSIs) have traditionally relied on rule-based systems like CIRCE, which require extensive manual maintenance and struggle with synonyms or paraphrases. We present an alternative approach based on semantic search with sentence-transformer embeddings, comparing naïve, disentangled, and centroid-based knowledge bases. Using more than 33,000 CIRCE-labelled queries and 12,400 synthetic queries, we evaluate various embedding models across different levels of the ATECO hierarchy. Results indicate that semantic search methods achieve over 80% top-5 matching at the 5-digit level, making them suitable for interactive coding tools. We further demonstrate that medium-sized models provide strong performance while remaining suitable for offline deployment in survey environments. These findings support transitioning from ATECO 2022 to ATECO 2025 and highlight the potential of embedding-based techniques to modernize automated coding in official statistics.

Keywords
Automated coding, Semantic search, Embedding models, Statistical Classifications

Introduction

Statistical classifications such as NACE, ISCO, PRODCOM, and their national variations, like ATECO in Italy, are essential tools for Official Statistics (OS). They provide a common language for describing economic and social phenomena, ensuring coherence and comparability across countries, different data sources, and statistical domains. With standardised categories, classifications enable the integration of survey data, administrative records, and business registers into consistent statistical frameworks, thereby supporting both national and international policymaking. Accurate classification is essential as misreporting can lead to biased estimates, reduced comparability, and inconsistencies in national accounts and business statistics. However, navigating these taxonomies can be challenging. Classifications are hierarchical and include many explanatory notes, which, while useful for domain experts, may confuse users without specialised knowledge. For enterprises and respondents filling out survey questionnaires or administrative forms, finding the correct code often requires substantial effort. This burden can lead to incorrect self-reporting and higher costs for both respondents and statistical offices.

To address these challenges, National Statistical Institutes (NSIs) have developed auxiliary systems to support classification tasks. A key example is the Italian Statistical Institute's (ISTAT) Comprehensive ISTAT R Coding Environment (CIRCE) system, a rule-based search engine designed to help retrieve economic activity codes (ATECO classification) from natural language queries (Istat, CIRCE, 2016). These systems have enhanced the quality of coding tasks and reduced respondent burden. Nevertheless, their deterministic design has significant limitations: rule sets need continuous manual updates whenever the classification is revised, and the lack of contextual

understanding limits their ability to generalize, especially with synonyms, complex phrasing, or typographical errors. These issues motivate the exploration of state-of-the-art semantic search techniques based on modern language models.

The rapid progress in Artificial Intelligence (AI), especially with the development of Transformer-based language models (Vaswani, et al., 2017), creates new opportunities to modernize coding processes. Unlike traditional rule-based systems, which rely on explicit keyword matching and manual rules, these models process and represent semantics. This allows them to interpret meaning beyond lexical matches and enables the development of search engines that can identify when different sentences convey similar meaning, handle synonyms and paraphrases, and understand queries even if they contain typos or unusual phrasing. As a result, Transformer-based methods significantly increase generalization capabilities and help systems manage the variability and ambiguity of natural language with unprecedented accuracy while considerably reducing the need for manual rule updates.

In this paper, we explore the use of semantic search techniques for coding economic activities. Building on ISTAT's experience with CIRCE, we propose an approach that utilizes Sentence Transformers (Reimers & Gurevych., 2019) to generate text embeddings of classification descriptions and user queries, enabling similarity-based retrieval of relevant codes. We compare our method with the CIRCE baseline, assessing the consistency of predictions across tens of thousands of real-world queries. Additionally, we utilize large language models (LLMs) to generate synthetic queries to improve the performance of our systems. The results demonstrate that semantic search methods can significantly improve the efficiency of coding processes while also enhancing generalization.

Our contribution is threefold:

1. We design and evaluate three alternative embedding-based knowledge bases for semantic search, ranging from naïve to centroid-based approaches.
2. We benchmark their performance against CIRCE using more than 33,000 real-world labelled queries, complemented with synthetic data generated by LLMs.
3. We demonstrate the potential of semantic search methods to reduce manual maintenance costs and increase robustness to synonyms, paraphrases, and typos, opening the way for the transition to the next ATECO revision.

The paper is structured as follows**:** Section 2 introduces the data, including the ATECO classification, the CIRCE-labelled queries, and the generated synthetic dataset; Section 3 describes the methodology and experimental setup, from traditional rule-based methods to semantic search and embedding models; Section 4 presents and analyses the results; Section 5 concludes with limitations and future research directions.

Finally, it is important to note that although ISTAT released an updated version of ATECO in April 2025 (hereafter referred to as ATECO 2025), this study focuses on ATECO 2007, with the 2022 revision (hereafter referred to as ATECO 2022), as the current CIRCE system and the associated labelled query dataset are based on this version. However, the proposed approach can directly extend to ATECO 2025, and future research will explicitly address this transition.

However, before proceeding, we believe it is appropriate to introduce Istat's CIRCE system.

CIRCE

NSIs have developed automated coding systems for decades. For instance, Automatic Coding by Text Recognition (ACTR) has been employed across a range of applications in the field of official statistics, including national population censuses (Tourigny & Moloney, 1995). The conceptual framework of the system draws upon a methodology initially developed by the United States Census

Bureau (Appel & Hellerman, 1983) and integrates text-matching algorithms subsequently refined by researchers at Statistics Canada (Wenzowski, 1988). To support activity coding, ISTAT developed CIRCE, a generalised software package designed to assign classification codes based on textual descriptions. Introduced as a successor to the ACTR v3 system, which ISTAT had used since 1998 but was no longer maintained, CIRCE was implemented in R to ensure portability across platforms and long-term sustainability. The system is classification- and language-independent, allowing for its application across various coding contexts. In practice, CIRCE has been most widely used for ATECO, the Italian version of NACE, supporting both businesses and internal statistical activities (Istat, ATECO 2022, 2022). CIRCE combines parsing routines with rule-based weighting algorithms to offer a reliable coding method, reducing manual work and ensuring greater consistency. However, each revision of the classification requires significant human effort to update CIRCE's rule sets, synonyms, and dictionaries.

The recent release of ATECO 2025 (Istat, ATECO 2025, 2025) illustrates this challenge: aligning CIRCE with the new version demands a substantial investment of staff time, making the process costly and difficult to scale. This revision has motivated the present research project, which aims to assess the potential of modern language models within the consolidated framework of ATECO 2022, and subsequently generalise the results to ATECO 2025.

Traditional rule-based systems for processing natural language, such as CIRCE, face structural limitations. They ensure transparency and consistency but struggle with synonyms, paraphrases, or unexpected linguistic variations, and they require extensive manual maintenance at every revision. These constraints provide the rationale for experimenting with semantic search methods based on embedding models and large language models, which aim to reduce manual updates while enhancing generalization and robustness.

Data

This section describes the sources used in our study and how they are prepared for experiments. We begin with the ATECO classification, then we introduce a large set of real-world queries collected through the CIRCE tool, labeled with 5-digit ATECO categories. This dataset is used to measure the alignment of our methodology with the rule-based baseline. Finally, we create a complementary set of synthetic queries with a large language model (LLM) to achieve broader lexical coverage at the most detailed level of the taxonomy (6-digit sub-categories) and to support centroid-based knowledge bases. Together, these sources enable us to evaluate both empirical performance on observed user data (CIRCE-labeled queries) and coverage/robustness in controlled settings (LLM-generated queries).

ATECO classification

The ATECO classification groups together economic activities that are assumed to be similar at various levels, respecting the principles of completeness and exhaustiveness. The individual categories are mutually exclusive, so each item must be classified in only one specific category, and they must be meaningful, in the sense that the breakdown must not fall below limits that could hinder its collection. The ATECO hierarchy is made of six different levels: (1) *Sections*, single-letter codes from A to U; (2) *Divisions*, 2-digit numeric codes from 01 to 99; (3) *Groups*, 3-digit numeric codes; (4) *Classes*, 4-digit numeric codes; (5) *Categories*, 5-digit numeric codes; (6) *Sub-categories*, 6-digit numeric codes. Table 1 shows a scheme for the hierarchy along the size of each level.

Table 1. The ATECO 2007 (rev. 2022) hierarchical structure.

	Section	Division	Group	Class	Category	Sub-category
Digits	1	2	3	4	5	6

Size	21	87	270	615	919	1240
Example	A, U	05, 73	46.7, 74.2	46.73	46.73.4	46.73.40

Beyond the numeric codes and titles, ATECO is enriched by explanatory notes that provide detailed guidance for coding (see Figure 1 for the NACE equivalent). These include:

- **Central notes**, which describe the general scope of a code.
- **Inclusion notes**, which provide examples of activities covered.
- **Exclusion notes**, which clarify boundaries by pointing to activities classified elsewhere.
- **Operational notes**, which support consistent interpretation in practice.

01.1 Growing of non-perennial crops
This group includes the growing of non-perennial crops, i.e. plants that do not last for more than two growing seasons. Included is the growing of these plants for the purpose of seed production.

01.11 Growing of cereals (except rice), leguminous crops and oil seeds
This class includes all forms of growing of cereals, leguminous crops and oil seeds in open fields. The growing of these crops is often combined within agricultural units.

This class includes:
- Growing of cereals such as:
  - Wheat
  - Grain maize
  - Sorghum
  - …

This class excludes:
- Growing of rice, see 01.12
- Growing of sweet corn, see 01.13
- …

Figure 1. Example of a NACE group (01.1) and class (01.11) along with inclusion notes (blue) and exclusion notes (orange). Both inclusion and exclusion notes located in class 01.11 apply to all nested categories and sub-categories. The text has been shortened for clarity purposes.

The combination of a hierarchical structure and detailed explanatory notes makes ATECO an exhaustive yet complex classification. While the notes enhance clarity and accuracy for human coders, they also make automated coding more difficult: an enterprise description may match multiple candidate codes at different hierarchical levels, and choosing the correct one often depends on subtle distinctions expressed in natural language. This inherent complexity is challenging to address with string-matching methods, but it is potentially beneficial for modern AI-based semantic search methods, where semantic richness often enhances model performance.

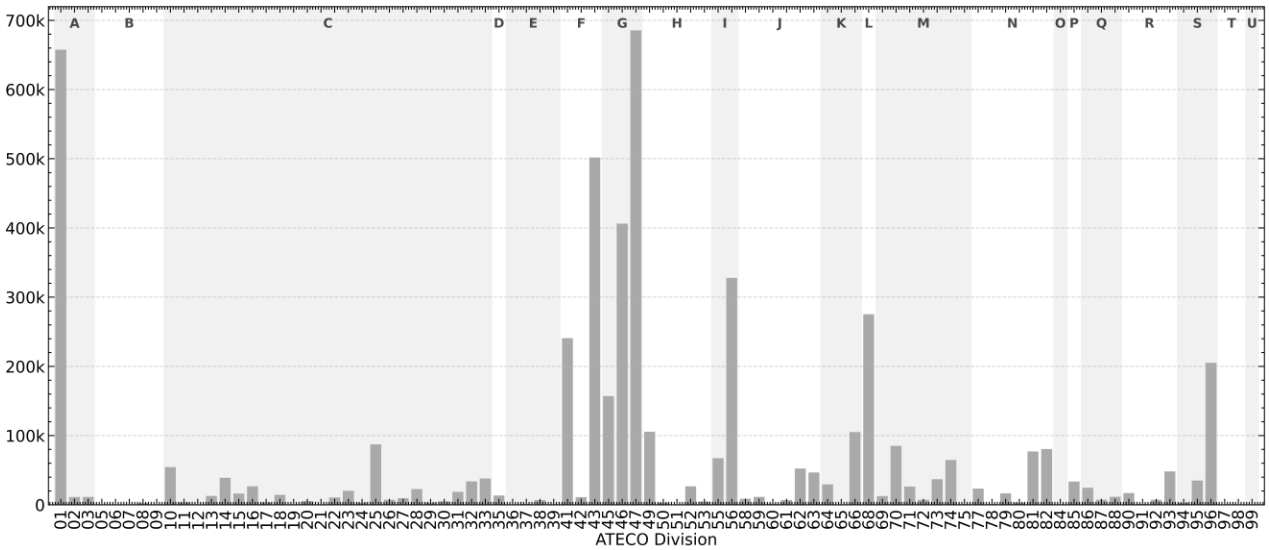

Figure 2. Distribution of active businesses in Italy by ATECO division (2-digit code) as of December 2024. The shaded background areas and capital letters indicate ATECO sections.

Figure 2 shows a strongly unbalanced distribution of enterprises across ATECO divisions. The largest division is 47, *retail trade*, with 685.462 active businesses, and the least popular is 99, *activities of extraterritorial organisations and bodies*, with only 5 active enterprises.

CIRCE-labelled real-world queries

The Italian national institute of statistics collected user queries made through the CIRCE-based platform, which helps businesses and professionals to find the right ATECO category (5-digit) starting from natural language descriptions.

We employ a total of 33,544 queries and run them through the CIRCE algorithm to extract predictions. Therefore, we obtain a CIRCE-labelled dataset, that assigns an ATECO category to each query. We will use this dataset to measure the alignment between our method and traditional rule-based methods. Figure 3 shows the distribution of CIRCE-assigned labels across ATECO divisions.

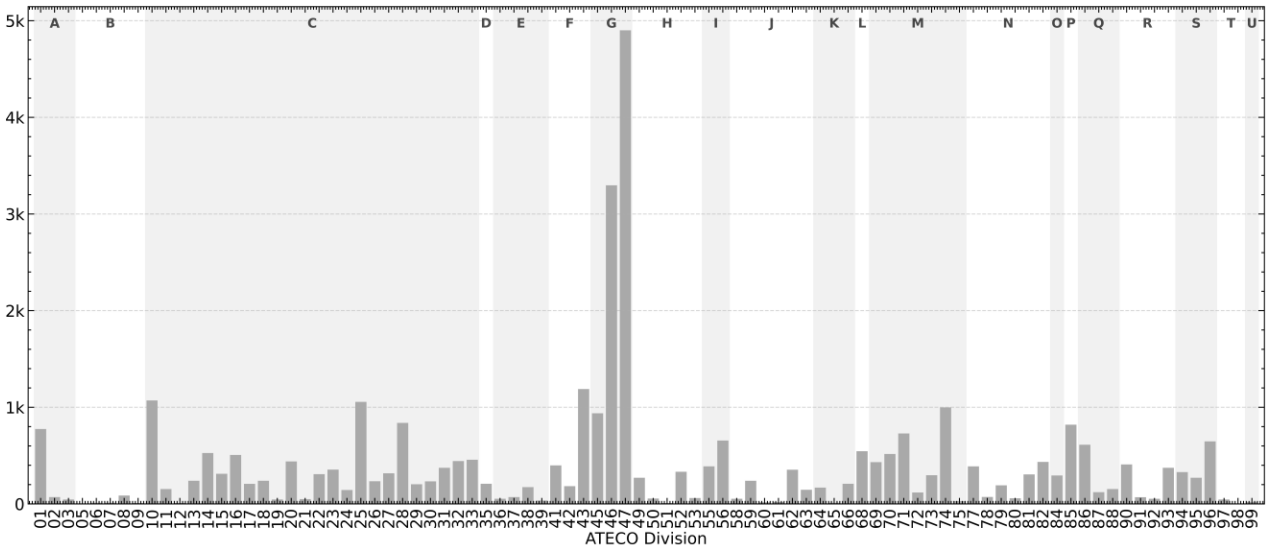

Figure 3. Distribution of CIRCE-assigned labels across ATECO divisions (2-digits). The shaded background areas and capital letters indicate ATECO sections.

We can see how the distribution of queries by division loosely resembles the distribution shown in Figure 2.

Synthetic queries

In addition to, real-world queries, we also generate a dataset of synthetic queries using a large language model (LLM), GPT-5-mini specifically (OpenAI, 2025). The model was instructed to create ten possible queries for each ATECO sub-category, resulting in a total of 12,400 synthetic queries. The system prompt used for the model is shown in Figure 4.

For each ATECO sub-category (6-digit), we create a rich textual description – which we refer to as *descriptor* – that encompasses all available information about a specific classification, including title, description, inclusion notes, exclusion notes, and hierarchy. We will explore this more in depth in the following paragraphs.

You are a generator of descriptions of economic activities carried out by businesses and professionals.

Here is what you need to do:

- You receive an economic classification as input: title + details (included and excluded activities, classification hierarchy. activities, classification hierarchy.
- You generate 10 examples of very short descriptions of activities carried out by enterprises, coherent with the classification.
- Focus on the specified activity. If present, be careful with the excluded activities.
- You do not generate text that is not one of the descriptions.

Important information: keep in mind that the descriptions that you generate serve as synthetic queries for a search engine, in which an enterprise/professional is asked to describe their activity with the aim of assigning a classification.

Here are some examples:

[8 examples of real queries here]

Format your output as a Python list.

Figure 4. System instructions for GPT-5-mini.

In addition to the system prompt, the LLM also receives the descriptor of an ATECO sub-category, which includes the most relevant information about that specific code. An example of such descriptor can be found in Figure 5.

Activity: Cultivation of other fruit trees, berries, and nuts.

This activity **includes**:

- Cultivation of berries: blueberries, currants, gooseberries, kiwis, raspberries, strawberries, other berries.
- Production of fruit seeds.
- Cultivation of edible nuts: almonds, cashews, hazelnuts, pistachios, walnuts, other nuts.
- Cultivation of carob, persimmons, pomegranates.
- Cultivation of other fruit trees or berry plants.

This activity **excludes**:

- Cultivation of coconuts, see 01.26.
- Gathering of wild berries and edible nuts, see 02.30.

Hierarchy: Crop and animal production, hunting and related service activities > Cultivation of permanent crops > Cultivation of other fruit trees, berries, and nuts.

Figure 5. Example of translated ATECO sub-category descriptor for code 01.25.00.

CIRCE assigns labels at the 5-digit category level, while our synthetic queries are generated at the 6-digit sub-category level. To make the two sources comparable in evaluation, we aggregate synthetic data from the 6-digit sub-categories to their 5-digit parent categories: specifically, for each 5-digit category we combine all synthetic queries originating from its 6-digit sub-categories and use their embeddings to compute a single 5-digit centroid for retrieval and scoring. All metrics reported against CIRCE labels are therefore calculated at the 5-digit granularity. The details of the centroid construction and similarity computation are provided in the Experimental design paragraph of the Methodology section.

Methodology

This section details our study's methodological framework. We begin by reviewing traditional rule-based methods such as CIRCE, which are still the standard in official statistics. Then, we introduce sentence-transformer models as advanced embedding techniques for semantic search. Finally, we outline the experimental design that allows for a systematic comparison of knowledge-base construction strategies and embedding models.

Traditional rule-based methods

Traditional approaches to automatic coding in official statistics are based on deterministic rule sets and weighted keyword matching. These methods rely on dictionaries derived from the classification, where each entry is pre-processed and indexed according to a user-defined parsing strategy. The parsing phase standardises the text by applying strategies such as stop-word removal, synonym replacement, or tokenisation, so that syntactically different descriptions with the same meaning can be aligned. The resulting dictionary entries are then represented as sets of weighted tokens (unigrams, bigrams, trigrams) that capture their *semantic signature*.

Once the dictionary is generated, free-text descriptions submitted by users undergo the same parsing procedure and are compared against all dictionary entries. CIRCE, based on the ACTR v3 engine, computes a similarity score S for each candidate, defined as a weighted average of matching terms. Depending on the maximum score S_1, the second-best score S_2, and the thresholds $(S_{\min}, S_{\max}, \Delta S)$, the algorithm produces four possible outcomes:

1. Unique, a single entry exceeds $S_{\max}$ and is clearly separated from the next best candidate.
2. Multiple, more than one entry exceeds $S_{\max}$ with scores too close to be distinguished.
3. Possible, the best score falls between $S_{\min}$ and $S_{\max}$.
4. Failed, no candidate exceeds $S_{\min}$.

This rule-based architecture has several strengths: it provides transparency, allows domain expert-driven tuning, and is effective when vocabularies and synonyms are comprehensively represented in the dictionary. At the same time, it also exhibits structural limitations:

- Parsing strategies maintenance, as tokenisation rules, normalisation, and synonym lists must be updated whenever classifications are revised.
- No semantic understanding, sentences are reduced to standardised token sets, which makes the approach fragile to paraphrases and long queries.
- Rigid matching logic, since exact or near-exact token match is required to trigger a match.
- Low scalability, as adapting to new domains, classifications or tasks requires regenerating dictionaries and strategies from scratch.

These limitations were particularly evident in the recent transition in Italy from ATECO 2007 (rev. 2022) to ATECO 2025, highlighting the significant cost in expert time required to maintain rule-based engines. They also motivate the exploration of language modelling-based approaches that can capture meaning and context more flexibly and reduce the reliance on hard-coded rules.

Sentence transformers

The whole premise of our proposed approach is to define a representation space for the economic activity taxonomy, which we refer to as *knowledge base*. In such setting, economic activity codes are represented as dense, high-dimensional vectors in the representation space (see Figure 6). To perform this encoding, we rely on sentence-level embedding models, i.e., sentence transformers (Reimers & Gurevych., 2019), which encode an entire text into a single embedding vector. Whether to encode each economic activity classification as a single text into a single vector or split it where possible into multiple texts is a design choice that we explore in the experimental design paragraph.

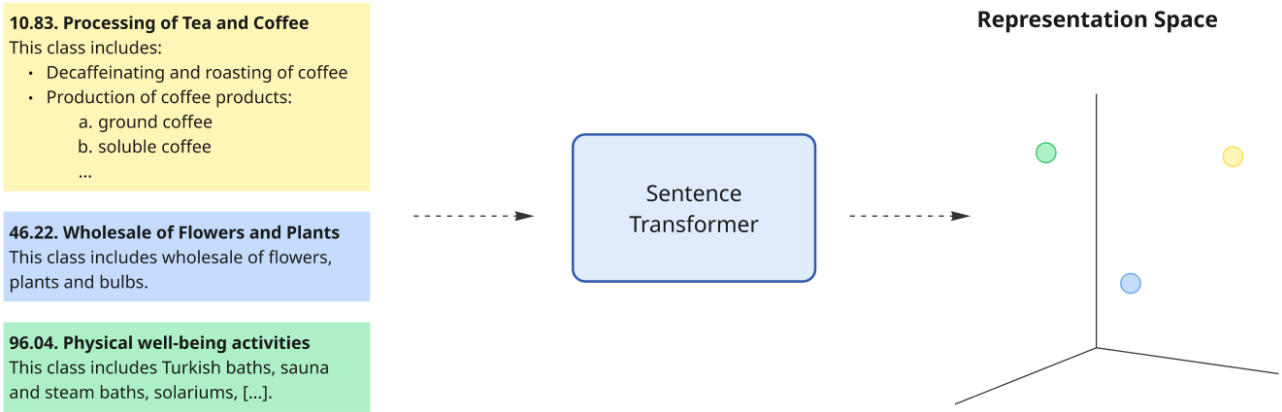

Figure 6. Each economic classification is mapped to a representation space via a sentence transformer model, where it is represented as a vector or a combination of vectors.

Throughout our analysis, we will utilize several sentence-level embedding models, as shown in Table 2. The models we test range from smaller sizes, around the 100 million parameters mark, to larger sizes, up to eight billion parameters.

Table 2. List of sentence transformer models along their number of parameters, the vector size of the embeddings they produce and the work that introduced them.

Name	Parameters	Vector size	Introduced by
Qwen-Embedding-8B	8,000 million	4,096	(Zhang, et al., 2025)
Qwen-Embedding-4B	4,000 million	2,560	(Zhang, et al., 2025)
Qwen-Embedding-0.6B	596 million	1,024	(Zhang, et al., 2025)
BGE-M3	568 million	1,024	(Chen, et al., 2024)
gte-multilingual-base	305 million	768	(Zhang, et al., 2024)
EmbeddingGemma	303 million	768	(Gemma, 2025)
paraphrase-multilingual	118 million	384	(Reimers & Gurevych., 2019)

Experimental design

There are several ways in which the creation of a knowledge base can be handled. Whether assigning multiple vectors to the same code or implementing a one-to-one relationship between codes and elements in the knowledge base, the principle is to create some kind of latent representation of the taxonomy in the embedding space.

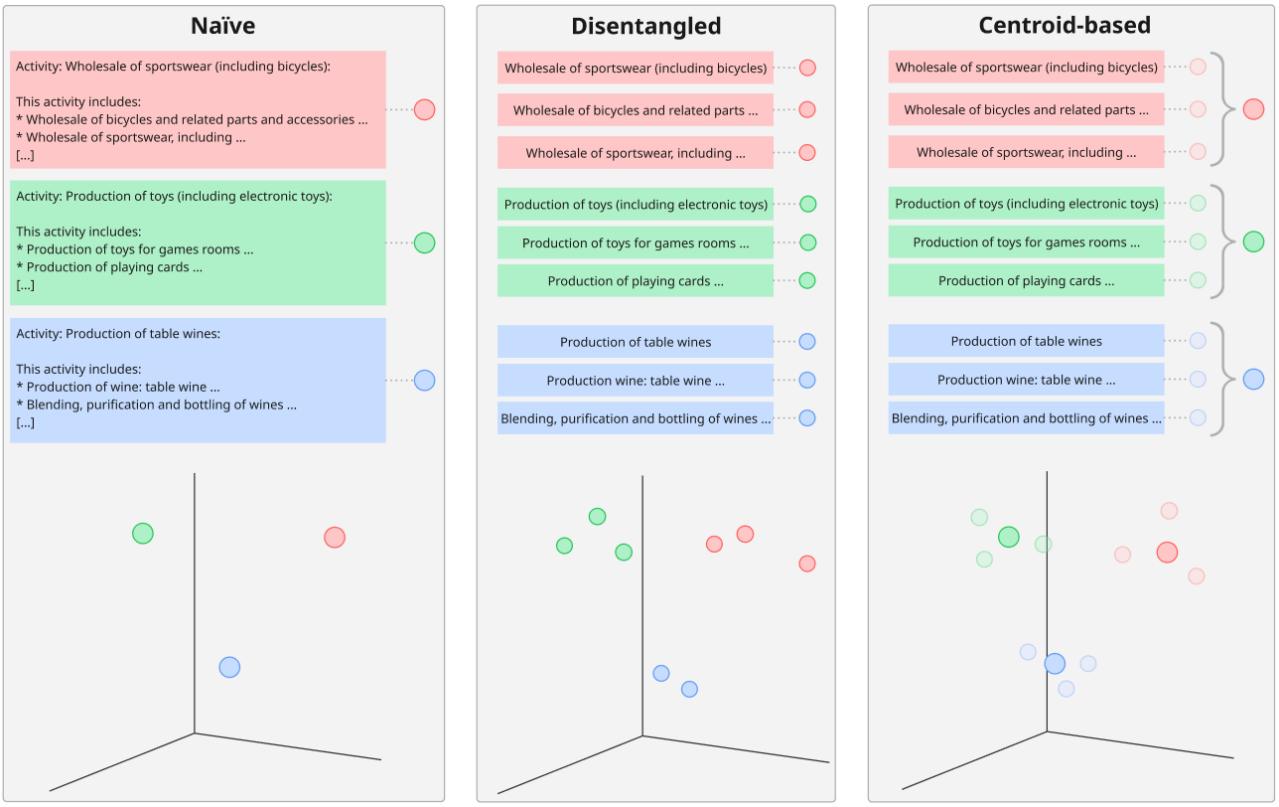

Figure 7. Illustration of the different encoding approaches. Circles represents embedding representation of texts, and different colours represent different ATECO categories.

Here, we explore three main ways to achieve such goal, illustrated in Figure 7. In particular:

- A naïve approach, where we group the information contained in the actual classification into a single textual description and encode it as a single vector.
- A disentangled approach, where we separate each classification into several sub-texts, whenever possible, for example by splitting lists into individual texts or by using synthetic queries. This way we obtain a one-to-many mapping in the knowledge base.
- A centroid-based approach, where we exploit the texts from the disentangled approach to create centroid embedding vectors to represent the entire category as the average of its own elements.

Formally, let $\pi: \mathcal{T} \to R^d$ be the sentence-embedding model that maps a text $t \in \mathcal{T}$ to a d-dimensional dense embedding vector. Let $\mathcal{C}$ be the set of economic activity codes, and for each code $c \in \mathcal{C}$ let $S_c = \{s_{c,1}, s_{c,2}, \dots, s_{c,n}\}$ be the set of textual elements, i.e. the pieces of text describing the code. Then:

Naïve approach (one-to-one) Choose a single, consolidated text $s_c^{\text{naïve}}$, the *descriptor*, for each code c. In this setting, the knowledge base stores a single vector per code, obtained as:

$$\boldsymbol{e}_c^{\text{naïve}} = \pi(s_c^{\text{naïve}}) \in R^d.$$

For a query q we compute its embedding vector $\boldsymbol{e_q} = \pi(q)$ and score code c via a measure of vector similarity (we use cosine similarity):

$$\text{score}_{\text{naïve}}(c|q) = \text{sim}\left(\boldsymbol{e}_q, \boldsymbol{e}_c^{\text{naïve}}\right).$$

At this point, we extract the top_k ATECO categories based on the computed similarities.

Disentangled approach (one-to-many) In this scenario, we keep separate embeddings for each sub-text of an ATECO category or synthetic query. For each $s_{c,j} \in \mathcal{S}_c$ compute:

$$\boldsymbol{e_{c,j}} = \pi(s_{c,j}) \in R^d, \quad (j = 1, \dots, n_c).$$

The knowledge base maps $c \mapsto \mathcal{E}_c = \{\boldsymbol{e_{c,1}}, \dots, \boldsymbol{e_{c,n_c}}\}$, where n_c is the code-specific number of sub-texts. At this point, we apply a max-pooling strategy to compute the score for each code:

$$\text{score}_{\text{dis}}^{\max}(c|q) = \max_{1 \le j \le n_c} \text{sim}\left(\boldsymbol{e_q}, \boldsymbol{e_{c,j}}\right).$$

Finally, we extract the top_k ATECO categories based on the computed similarities.

Centroid-based approach (one-to-one) Given a set of texts $\mathcal{S}_c = \{s_{c,1}, \dots, s_{c,m_c}\}$ where m_c is the category-specific number of synthetic queries—which can vary as both texts from the ATECO classification and synthetic queries are obtained for sub-categories and aggregated at the category level—compute the category centroid embedding as:

$$\bar{\boldsymbol{e}}_c = \frac{1}{m_c} \sum_{j=1}^{m_c} \pi(s_{c,j}) \in R^d.$$

Then, compute the scores exactly as in the naïve approach:

$$\text{score}_{\text{cen}}(c|q) = \text{sim}(\pi(q), \bar{\boldsymbol{e}}_c).$$

Again, the top_k ATECO categories are extracted according to their relative similarity scores.

We measure the performance of our approach analysing the matching against CIRCE-based predictions, and we do it at two hierarchical levels. The division level, 2-digit ATECO codes, and the category level, 5-digit ATECO codes. To obtain metrics at the division level, we simply extract the first two digits from the retrieved ATECO categories.

In the disentangled and centroid-based approaches, we run the procedure both with texts extracted from the official ATECO classification and synthetic queries.

Results

In this section we first present the results obtained with different embedding models across different encoding and classification approaches. We measure as the average matching between CIRCE-assigned labels and semantic search-assigned labels.

The overall results are shown in Table 3, which presents matching at different classification hierarchy levels (ATECO division and category), and at different ranks. In particular, match@1, scored if there is a match between the CIRCE label and the most similar semantic search label, match@3, scored if the CIRCE label is in the top 3 most similar semantic search labels, and match@5, scored if the CIRCE label is in the top 5 most similar semantic search labels.

Table 3. Matching at different ranks, across levels, models, and approaches.

Approach	Model	Division (2-digit)			Category (5-digit)		
		Match@1	Match@3	Match@5	Match@1	Match@3	Match@5
Naïve	Qwen-Embedding-8B	0.6116	0.8132	0.8657	0.4188	0.6394	0.7184
	Qwen-Embedding-4B	**0.6976**	**0.8650**	**0.9094**	**0.5281**	**0.7397**	**0.8070**
	Qwen-Embedding-0.6B	0.3780	0.5596	0.6432	0.2309	0.3718	0.4578
	BGE-M3	0.6586	0.8114	0.8608	0.4694	0.6619	0.7338
	gte-multilingual-base	0.6020	0.7877	0.8444	0.3998	0.6010	0.6827
	EmbeddingGemma	0.0418	0.0803	0.1173	0.0050	0.0097	0.0140
	paraphrase-multilingual	0.4566	0.6404	0.7094	0.2465	0.4107	0.4841
Disentangled (official)	Qwen-Embedding-8B	**0.6602**	0.8401	0.8955	0.4706	0.6553	0.7224
	Qwen-Embedding-4B	0.6413	0.8307	0.8896	0.4518	0.6358	0.7109
	Qwen-Embedding-0.6B	0.5719	0.7601	0.8243	0.3778	0.5410	0.6096
	BGE-M3	0.6214	0.8102	0.8664	0.4327	0.6337	0.7112
	gte-multilingual-base	0.6435	**0.8424**	**0.8957**	**0.4844**	**0.6929**	**0.7671**
	EmbeddingGemma	0.3302	0.4380	0.4964	0.1750	0.2513	0.2903
	paraphrase-multilingual	0.5428	0.7280	0.7913	0.3542	0.5294	0.6017
Disentangled (synthetic)	Qwen-Embedding-8B	**0.6438**	**0.8415**	**0.8998**	**0.4462**	**0.6800**	**0.7657**
	Qwen-Embedding-4B	0.6195	0.8368	0.8993	0.4269	0.6638	0.7569
	Qwen-Embedding-0.6B	0.5447	0.7684	0.8417	0.3398	0.5568	0.6468
	BGE-M3	0.5986	0.8057	0.8721	0.3880	0.6189	0.7094
	gte-multilingual-base	0.5378	0.7775	0.8540	0.3488	0.5809	0.6799
	EmbeddingGemma	0.3235	0.4848	0.5594	0.1328	0.2324	0.2859
	paraphrase-multilingual	0.4189	0.6563	0.7503	0.2234	0.4047	0.5018
Centroid (official)	Qwen-Embedding-8B	**0.6817**	**0.8447**	**0.8913**	0.4886	0.6773	0.7427
	Qwen-Embedding-4B	0.6632	0.8329	0.8833	0.4683	0.6584	0.7267
	Qwen-Embedding-0.6B	0.5882	0.7572	0.8145	0.3822	0.5472	0.6159
	BGE-M3	0.6633	0.8233	0.8691	0.4695	0.6635	0.7350
	gte-multilingual-base	0.6587	0.8396	0.8881	**0.4894**	**0.7008**	**0.7684**
	EmbeddingGemma	0.3244	0.4358	0.4933	0.1700	0.2510	0.2918
	paraphrase-multilingual	0.5469	0.7276	0.7897	0.3522	0.5382	0.6118
Centroid (synthetic)	Qwen-Embedding-8B	**0.7143**	**0.8708**	**0.9139**	**0.5455**	**0.7513**	**0.8215**
	Qwen-Embedding-4B	0.7093	0.8696	0.9138	0.5351	0.7490	0.8207
	Qwen-Embedding-0.6B	0.6164	0.7896	0.8467	0.4199	0.6245	0.7042
	BGE-M3	0.6840	0.8385	0.8831	0.4820	0.6937	0.7689
	gte-multilingual-base	0.6338	0.8198	0.8708	0.4533	0.6719	0.7490
	EmbeddingGemma	0.3841	0.5356	0.6076	0.2047	0.3263	0.3841
	paraphrase-multilingual	0.5014	0.7012	0.7732	0.3050	0.4992	0.5805

As we can see, the best overall performance is reached, perhaps unsurprisingly, by the largest model, *Qwen-Embedding-8B*, with over 82% match@5 for 5-digit category and over 91% match@5 for 2-digit divisions. However, the large size of this model makes it impractical for deployment in a production environment. In general, the naïve approach seems sub-optimal, especially for smaller models, while disentangled and centroid-based approaches look more promising. The effect of synthetic queries over official ATECO notes is ambiguous, as it improves the performance for some models, e.g. *BGE-M3*, while is detrimental for other ones, e.g. *gte-multilingual-base*.

Taking into consideration (a) the size of the models, and therefore the feasibility of their usage in a production environment, and (b) the overall performance, the most promising scenario seems to be the centroid-based approaches using the *gte-multilingual-base* and *BGE-M3* models. From this moment on, we will focus our analysis on this scenario.

Matching by division

In this paragraph, we will inspect the average matching (match@5) by 2-digit ATECO divisions. We do this to check whether our models systematically underperform on some divisions, or if the performance is smooth across the classification.

We computed this division-wise matching for the *gte-multilingual-base* model using the centroid-based approach with official sub-texts, and the results are shown in Figure 8.

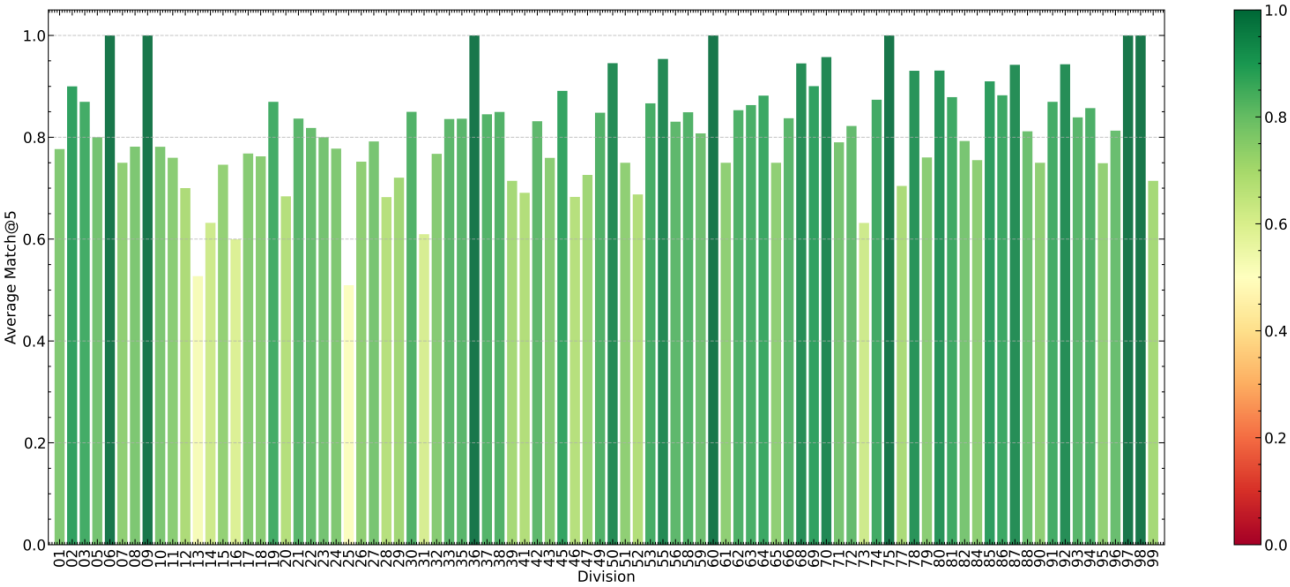

Figure 8. Average division-wise match rate (top 5) between CIRCE and semantic approaches

As we can see, none of the divisions show matching values under 50%. The worst performance is attained at division 13, “textile industries”, and 25, “manufacture of metal products (excluding machinery and equipment)”. For division 13, the most frequently assigned division in mismatches is 17, “manufacture of paper and paper products”, while for division 25, it is 24, “metallurgy”.

However, the underperformance does not seem to be consistent across models. For example, *BGE-M3* severely underperforms on division 43, “specialised construction work”, obtaining a match@5 of less than 40%, while *gte-multilingual-base* almost reaches an 80% match@5 score. This inconsistency is likely due to the way these models are trained, specifically to the dataset employed during training. That is why there is no one-fits-all solution as the performance of embedding models highly depends on the task at hand.

Similarity thresholds

All the results shown up until this point are obtained without setting any threshold on the cosine similarity between query embeddings and knowledge base embeddings to assign a code. This means that whatever the similarity, we always assign an ATECO code, even if the value is low.

However, we might want to set a threshold, so that if the query is not semantically similar enough, we do not assign any label to it. Figure 9 shows how the top 1-matching at the ATECO category level (5-digit) varies if we apply a similarity threshold, starting from a value of 0.5 up until 0.9 for two of the analysed models, *gte-multilingual-base* and *BGE-M3*. The figure shows results for the centroid-based approach using official sub-texts.

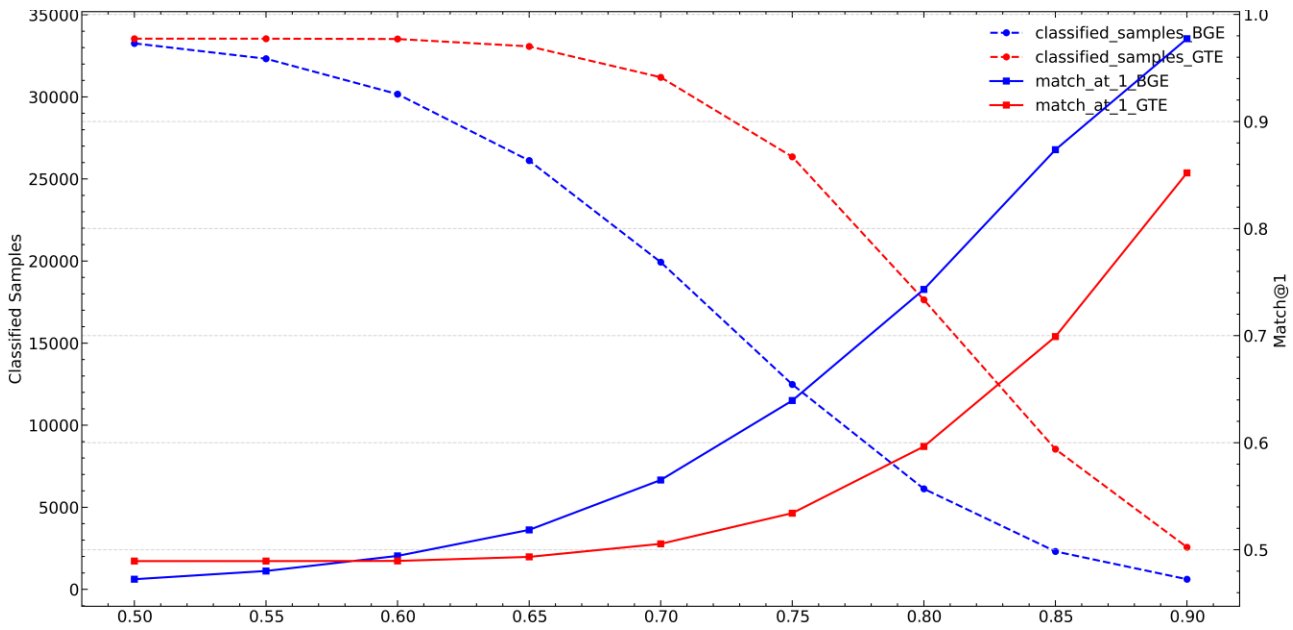

Figure 9. Matching (top 1) between CIRCE and semantic approaches at different levels of similarity thresholds using the *gte-multilingual-base* (red) and *BGE-M3* (blue) models. The matchings are shown as whole lines, while dashed lines show how many queries have been classified.

The first thing we notice, is that *BGE-M3* shows a much steeper increase in matching values and decrease in classified samples (queries with a top 1 similarity value over the threshold) compared to *gte-multilingual-base*. However, setting thresholds on cosine similarity can lead to unexpected results, as some texts can score systematically lower or higher similarities depending on how the model was trained. We leave the exploration of this matter to future research and further discuss it in the concluding section.

Semantic richness and performance

Traditional rule-based methods are mainly based on keyword matching. That is, if a query only contains a word that is specific enough the system can easily classify it. However, methods based on embedding models usually benefit from higher levels of verbosity in descriptions.

Table 4. Matching at different ranks, across levels, models, and word filter counts. The results shown are from the centroid (official) approach.

Model	Word count filter	Division (2-digit)			Category (5-digit)		
		Match@1	Match@3	Match@5	Match@1	Match@3	Match@5
BGE-M3	Standard	0.6633	0.8233	0.8691	0.4695	0.6635	0.7350
	> 4	**0.7643**	0.8950	0.9264	**0.5512**	0.7425	0.8070
gte-multilingual-base	Standard	0.6587	0.8396	0.8881	0.4894	0.7008	0.7684
	> 4	0.7270	**0.9021**	**0.9394**	0.5419	**0.7554**	**0.8185**

To show this, we filter the queries for texts containing at least 5 words. This leaves us with 12,518 queries out of the original 33,544. Classifying this semantically richer subset leads to a consistent improvement in matching. Table 4 shows this improvement for the *BGE-M3* and *gte-multilingual-base* models in the centroid-based approach using official ATECO sub-texts. The models achieve higher levels of matching across the board, with a 5 to 10 percentage points improvement compared to their performance on the whole dataset.

A note on mismatches

While CIRCE's large ruleset leads to very precise classification, the system can make mistakes. Therefore, before taking the CIRCE-assigned labels as our gold standard, we checked a sample of

mismatches with the *gte-multilingual-base* model to assess whether our model was misclassifying the query or if there was some ambiguity or a CIRCE classification error.

The sample was extracted from the subset of queries where CIRCE and semantic methods assigned mismatching codes at the 2-digit division level, i.e. "strong" mismatches. Figure 10 shows some random examples from this sample.

Query: "car accessories"
CIRCE: 45.32.0 – Retail trade of car parts and accessories.
Semantic: 29.32.0 – Manufacture of other parts and accessories for cars.
Ambiguous

Query: "shipboard and naval supplies"
CIRCE: 56.29.2 – Ongoing catering on a contractual basis.
Semantic: 30.11.0 – Shipyards for metal and non-metal constructions.
Semantic

Query: "brand management"
CIRCE: 77.40.0 – Granting of rights to exploit intellectual property and similar products […]
Semantic: 73.11.0 – Advertising agencies.
Ambiguous

Query: "production and sale of rubber and plastic"
CIRCE: 22.29.0 – Manufacture of other plastic products.
Semantic: 28.96.0 – Manufacture of machinery for the plastics and rubber industry […]
CIRCE

Query: "wedding video editing"
CIRCE: 74.20.2 – Photographic laboratories for developing and printing.
Semantic: 43.29.0 – Other construction and installation work.
CIRCE

Figure 10. Examples of mismatches along with top 1 predictions. Coloured text shows our interpretation of the mismatch: *ambiguous* if we couldn't tell with certainty whose prediction was correct, *semantic* if the semantic approach improved the classification, and *CIRCE* if CIRCE's label was more fitting.

In general, we noticed a large number of ambiguous classifications, especially for unclear queries, i.e., queries that can lead to multiple codes. If we look at the last example in Figure 9, while the semantic search label is clearly wrong, the third-assigned label, with only a 0.013 difference in similarity, is "74.20.1 – photography activities", which is arguably a more fitting category compared to the CIRCE-assigned one.

Conclusion

This study examined the use of semantic search methods for the automated coding of economic activities, using the Italian ATECO classification as a case study. By comparing different knowledge-base construction strategies and embedding models against more than 33,000 CIRCE-labelled queries, we demonstrated that modern language model approaches can achieve high coding accuracy while reducing the need for expensive manual maintenance. Disentangled and centroid-based approaches strike a balance between performance and model size, with the CIRCE-assigned category appearing in the top five retrieved codes in over 76% of the cases using medium-size embedding models and over 82% of the cases using large embedding models, while this percentages rise to around 90% if we consider 2-digit codes, and to 94% if we consider semantically rich queries. Such performance already meets the requirements of practical coding tools, where

users typically choose the most suitable label from a shortlist rather than relying on a single prediction.

A second key finding concerns the feasibility of adopting medium-sized models. While very large models such as Qwen-8B achieve the highest match@1 scores, models in the 300–600 million parameter range (e.g., *gte-multilingual-base* and *BGE-M3*) offer a favourable balance between accuracy and computational cost, as they can sustain reliable interactive coding systems without the need for heavy infrastructure. This aspect is particularly relevant for official statistics: medium-sized models enable scenarios in which semantic search engines can be deployed offline, for instance, during CAWI surveys. In such a scenario, interviewers or respondents could query the classification directly from their device, without internet access, and receive a shortlist of suggested codes. This would enhance the accuracy of self-classification and the quality of data collected in the field, while also reducing respondent burden and the need for post-survey editing by statistical offices.

Arguably the most significant advantage of semantic search approaches is that they do not rely on any human-annotated ruleset, reducing drastically the need for manual work. In this case study, we obtained some of the best results solely relying on the official ATECO notes, and the generation of synthetic queries is still significantly more cost-efficient compared to manual annotation, even including rigorous validation steps.

Another important element in our analysis is the role of similarity thresholds. Our experiments show that increasing thresholds can improve performance but decrease the share of classified queries, with different models presenting different trade-offs. Future work could explore alternative approaches to cosine similarity-based retrieval, for instance linear discriminant analysis (LDA) over the centroid embeddings, which would provide classification probabilities which can be used to filter plausible codes, or conformal prediction (Vovk, Gammerman, & Shafer, 2005), which could be used to create statistically valid prediction sets. However, the latter would need an extensive human-labelled dataset.

The methods proposed in this paper provide a path for the transition to the upcoming ATECO 2025 revision and can easily be adapted to other statistical classifications such as ISCO or PRODCOM. Future work will focus on expert validation of predictions, embedding model fine-tuning, the integration of threshold-based decision outcomes, and the development of balanced evaluation datasets. Embedding-based search engines represent a promising step toward more efficient and generalisable coding tools in official statistics.

All data and code needed for reproducibility is publicly available on GitHub (istat-methodology/semantic-auto-classification) except for CIRCE queries, which can contain sensitive information. However, to facilitate the reproducibility of our work, we published the sole embeddings for CIRCE queries, so that our work can be fully reproduced.

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